INCH-POUND
MIL-PRF-32760
22 February 2024

MILITARY PERFORMANCE SPECIFICATION

STAND, TEST, AIRCRAFT GENERATOR AND CONSTANT SPEED DRIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the aircraft generator test stand (AGTS). The AGTS replaces, and consolidates obsolete assets of the 400 cycle, alternating current (AC) aircraft generator and constant speed drive test stand, (Legacy K400B, K738XL/K400B-XL, MC-2000).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to AFLCMC/EZSS, 2145 Monahan Way, Bldg. 28, and Wright-Patterson AFB, OH, 45433-7017 or emailed to Engineering.Standards@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

TT-C-490	Chemical Conversion Coatings and Pretreatments for Metallic Substrates (Base for Organic Coatings)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-PRF-7808	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-PRF-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-DTL-16034	Detail Specification: Meters, Electrical-Indicating (Switchboard and Portable Types), General Specification for
MIL-PRF-21480	Performance Specification: Generator System, Electric Power, 400 Hertz, Alternating Current, Aircraft, General Specification for
MIL-PRF-23377	Primer Coatings: Epoxy, High-Solids
MIL-PRF-32550	Metal-Rich Primer
MIL-DTL-53022	Primer, Epoxy Coating, Corrosion Inhibiting Lead and Chromate Free
MIL-DTL-53030	Primer Coating, Epoxy, Water Reducible, Lead and Chromate Free
MIL-DTL-81706	Chemical Conversion Materials for Coating Aluminum and Aluminum Alloys
MIL-PRF-85285	Coating: Polyurethane, Aircraft and Support Equipment

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-461	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-810	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-882	System Safety
MIL-STD-889	Dissimilar Metals

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MIL-STD-1472 Human Engineering

DEPARTMENT OF DEFENSE HANDBOOK

MIL-HDBK-470 Designing and Developing Maintainable Products and Systems

(Copies of these documents are available online at https://assist.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

40CFR82 Protection of Stratospheric Ozone

(Copies of this document are available online at https://www.ecfr.gov/.)

DEPARTMENT OF DEFENSE INSTRUCTIONS

DODI 8500.01 Cybersecurity

DODI 8510.01 Risk Management Framework for DoD Systems

(Copies of these documents are available online at (https://www.esd.whs.mil/dd/dod-issuances/.)

AIR FORCE INSTRUCTIONS (AFI)

AFI 17-101 Risk Management Framework for Air Force Information Technology

(Copies of this document are available online at https://www.e-publishing.af.mil/product-index/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

THE AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

B18.3-2012 Socket Cap, Shoulder, Set Screws, and Hex Keys (Inch Series)

(Copies of this document is available online at https://www.asme.org.)

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ASTM INTERNATIONAL

F835-20	Standard Specification for Alloy Steel Socket Button and Flat Countersunk Head Cap Screws
F1136/F1136M-11	Standard Specification for Zinc/Aluminum Corrosion Protective Coatings for Fasteners

(Copies of this document is available online at https://www.astm.org/.)

NACE INTERNATIONAL

SSPC-SP10/NACE No.2	Near-White Metal Blast Cleaning
SSPC-Vis 1	Guide and Reference Photographs for Steel Surfaces Prepared by Dry Abrasive Blast Cleaning

(Copies of these documents are available online at https://www.nace.org/.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	National Electrical Code, 2020

(Copies of this document are available online at https://www.nfpa.org/.)

SAE INTERNATIONAL

AMS-STD-595	Colors Used in Government Procurement
J429-2014	Mechanical and Material Requirements for Externally Threaded Fasteners
J995-2017	Mechanical and Material Requirements for Steel Nuts

(Copies of these documents are available online at https://www.sae.org/.)

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST SP 800-53	Security and Privacy Controls for Federal Information Systems and Organizations

(Copies of this document are available online at https://www.nist.gov/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Preproduction. When specified (see 6.2), one sample shall be subjected to preproduction inspection in accordance with 4.2.

3.2 Description and general layout. The aircraft generator test stand (AGTS) replaces obsolete test stands (legacy K400B, K738XL/K400B-XL, MC-2000) and it shall provide end user testing capabilities to field level technicians for the following Aircraft Mission-Design Series (MDS): A-10, F-15C, F-15E, F-16, B-1 & F-15EX. The AGTS shall consist of:

a. Electric AC/DC motor and gearbox

b. Hydraulic Supply

c. Power Supply/Drive Controller and Console

d. Resistive/Reactive Load Bank as part of the test station. Preconfigured test station layout is determined by manufacturer.

3.3 Performance. Provide benchmarking/testing of power generation assets for the following MDS: A-10, F-15C, F-15E, F-16, B-1 & F-15EX. The AGTS shall be equipped with an appropriately sized electric motor to provide rotation characteristics of running aircraft engines, mechanical power application systems, and applicable adapters to facilitate mechanical connection between test stand and Units Under Test (UUT).

3.3.1 AGTS system. The AGTS shall be equipped with an electric motor to provide rotation characteristics of running aircraft engines, mechanical power application systems, and applicable adapters to facilitate mechanical connection between test stand and UUT.

a. 460V Line to Line ±5%, 3-Phase, 60 hertz (Hz) (420 Amps continuous, 630A/1 minute)

b. 380, 400, or 415V Line to Line ±5%, 3-Phase, 50 Hz (930 Amps continuous, 1400A/1 minute)

3.3.1.1 Mechanical power/motor system. Drive Stand consists of an Electric motor and Gearbox. Gearbox provides a mounting pad with a splined output shaft required for use in the testing of generators and Constant Speed Drives (CSD).

a. Motor: 300 horsepower (HP) minimum, drip-proof, guarded, splash-proof, 104 degrees Fahrenheit (deg F) ambient operating temperature, continuous duty, clockwise (CW) or counter-clockwise (CCW) rotation.

b. Gearbox: 15000 revolutions per minimum (RPM) minimum output, adapter to accommodate CSD. (Part Number: 739827B or equivalent)

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3.3.1.1.1 Hydraulic supply. The hydraulic supply system shall consist of a heat exchanger, supply pump, scavenge pump, reservoir, filter, and heater. Acceptable for use with MIL-PRF-7808 or MIL-L-23699 lubrication oil in accordance with technical order (T.O.) 9H6-5-14-2.

3.3.1.1.2 Heat exchanger. High efficiency heat exchanger for regulating thermal values within the hydraulic system. Nominal capacity shall have a minimum of 300,000 British thermal units per hour (BTU/hr).

3.3.1.1.3 Supply pump. A pump shall be provided for hydraulic supply. The pump shall have a minimum capacity of 12 gallons per minute (GPM).

3.3.1.1.4 Scavenge pump. A scavenge pump shall be provided for return hydraulics. The pump shall have a minimum capacity of 12 GPM.

3.3.1.1.5 Reservoir. A reservoir shall be provided for the hydraulic system. The reservoir shall have the minimum capacity of 15 gallons.

3.3.1.1.6 Filter. A filter shall be fitted to be end user changeable. The filter shall have a minimum of 12 micron.

3.3.1.1.7 Heater. The heater shall be an open loop heater for regulating thermal values within the hydraulic system. The heater shall be 12 kilowatts (kW), open loop.

3.3.1.1.8 Power supply/drive controller. Input power shall be supplied with input power disconnect.

a. Minimum input option 1: 380/400/415 volts line to line (V L-L), 50 Hz ±5% Line regulation, taps provided for ±2.5% and ±5% voltage correction.

b. Minimum input option 2: 460V L-L, ±5%, 3-Phase, 60 Hz ±5% Line regulation, taps provided for ±2.5% and ±5% voltage correction.

3.3.1.1.9 Resistive/reactive load bank. A load bank shall be applying a load value to power generation with 150% and 200% overload capability, the load bank shall be rated under full load at 438,000 BTU/hr or better. 123kW (93kVAR) minimum, 115/200VAC nominal, 130/225VAC maximum, Generator Load (400-2000Hz), permanent magnet generator (PMG) load (800-5600Hz). Load resolution: (3-Phase balanced) 1 kW (1-phase trim) 33 Watts.

3.3.1.1.10 Flushing. The AGTS shall be able to perform aircraft generator flushing capabilities.

3.3.2 Control system. The AGTS shall be designed to control, indicate, monitor, alarm, and display the functions listed below. Drive speed input (minimum 100 RPM/sec) drive thermal regulation, A/C voltage and current metering, D/C voltage and current metering, frequency output, flow metering, pressure metering and temperature metering.

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TABLE I. Requirements.

Minimum metering values (All specifications valid for frequencies between 350 and 2300 Hz)	
A/C Voltage	
Generator and PMG Phase A, B, C	(5-450V True RMS ±0.1 volts, ±0.12% reading)
A/C Current	
Generator Phase A, B, C	0-1000 true rms amperes ±1 ampere (±0.50% reading, ±0.07% range)
PMG Phase A, B, C	0-50 true rms amperes ±0.01 ampere (±0.50% reading, ±0.07% range)
D/C Voltage	
Field Voltage	0-200VDC, ±0.01 volts, ±0.75% reading, ±0.25% range)
D/C Current	
Field Current	0-20 amperes, ±0.001 amperes, ±0.75% reading, ±0.2% range)
Frequency	
Generator/PMG Frequency	350-5 600Hz ±0.1Hz
Watts	
Generator Phase A, B, C	0-200 KW, ±0.1 KW, ±0.5% reading, ±0.1% range
Flow Metering	
Oil Flow	1-20 GPM ±0.01 GPM, ±0.5% reading, ±0.5% range
Pressure Metering	
Inlet/Return pressure	0-400 PSIG ±0.1 PSIG, ±0.25% reading, ±0.5% range
Charge pressure	0-600 PSIG ±0.1 PSIG, ±0.25% reading, ±0.5% range
Case pressure	-14 to 35 PSIG ±0.1 PSIG, ±0.25% reading, ±0.5% range
Temperature Metering	
Inlet/Return Oil Temperatures	50-400 °F ±1 °F, ±2 °F reading

3.4 Environmental conditions.

3.4.1 Operating temperature range. The ATGS shall be capable of operating in ambient temperatures ranging from 50 to 130 °F.

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3.4.2 Storage temperature range. The ATGS shall be capable of being stored in ambient temperatures ranging from -65 to 130 °F.

3.5 Weight and dimensions. The size and weight of the ATGS shall be as small as practicable consistent with the requirements specified herein. Maximum size and weight shall not exceed:

TABLE II. Maximum weight and dimensions.

Weight	28000 pounds.
Length	160 inches.
Width	350 inches.
Height	100 inches

3.6 Transportability. The AGTS shall be transportable on aircraft, truck or trailer, railcar, and ship and shall be capable of withstanding the mechanical shock, vibration, temperature, humidity, and pressure characteristics of these modes of transportation.

3.7 Reliability, maintainability, and service life.

3.7.1 Reliability. The AGTS shall have a mean time between failure (MTBF) of at least 50 hours measured at 95 percent one-sided confidence limits. Definitions of reliability terms shall be in accordance with A.3.

3.7.2 Maintainability. The AGTS shall be designed for maintainability in accordance with 5.9 through 5.9.22.9 of MIL-STD-1472; forces shall not exceed those specified for both males and females.

3.7.2.1 Preventive maintenance. The recommended routine preventive maintenance interval (PMI) shall be at least 500 operating hours. Preventive maintenance tasks, other than tasks required daily or before operation, shall not require more than 100 man-hours to complete. This shall be calculated by:

$$PMMH_T = \sum_{i=1}^{N} \frac{PMMH_i}{i}$$

where:

$PMMH_T$ is the total preventive maintenance man-hours,

i is a multiple of the routine PMI (that is, i=1 for the routine PMI, i=2 for every other routine PMI), and

$PMMH_i$ is the preventive maintenance man-hours at the ith PMI.

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3.7.3 Service life. The AGTS shall be designed for a minimum service life of 20 years/10,000 hours, without a need for major overhaul or repair. Service life does not apply to wear items such as filters, electric motor brushes, fuses, hoses, and cables.

3.7.4 Warranty. The AGTS shall have a warranty period of one year after installation, or 18 months after shipment, whichever occurs first.

3.8 Human systems integration (HSI). The AGTS shall be designed in accordance with MIL-STD-1472 for ease of operation, inspection, and maintenance.

3.9 Electromagnetic interference (EMI). The ATGS shall be in accordance with the following radiated emission and susceptibility requirements of MIL-STD-461: RE102 and RS103.

3.10 Cybersecurity and Software. The AGTS shall be designed with cybersecurity considerations IAW DODI 8500.01, DODI 8510.01, and AFI 17-101 along with referenced security controls for applicable Information Technology (IT) and Operational Technology (OT), commonly referred to as Platform Information Technology (PIT), and all associated software and firmware. The AGTS shall implement all applicable security controls from National Institute of Standards and Technology (NIST) Special Publication (SP) 800-53 as allocated through the IT categorization and control selection process of Risk Management Framework (RMF).

3.10.1 Operational Software. Any operating or system software, application software, and controller firmware shall, at a minimum, implement the following technical security controls to the maximum extent practical:

a. Configured in accordance with the appropriate Defense Information Systems Agency (DISA) Security Technical Implementation Guide (STIG) as applicable.
b. Utilize Role Based Access Control (RBAC) with basic user account(s) and administrator account(s) with elevated privileges. All user level functions required to operate the AGTS shall be installed to and executed from a user level account without the need for elevated privileges.
c. Restrict actions from being performed without user identification and authentication, except for those actions necessary to ensure safety and prevent mishaps.
d. Configured to disable ports, protocols, and services which are not required for the system to perform its expected functions.
e. Capable of verifying integrity of application software and firmware prior to installation or operational use.
f. Utilize antivirus protection as applicable.
g. Software shall not have any high-level vulnerabilities as reported in the National Institute of Standards and Technology (NIST) National Vulnerability Database (https://nvd.nist.gov/vuln) concerning Common Vulnerability and Exposures (CVEs).

3.10.2 Diagnostic software. The AGTS shall have internal diagnostic capability to identify and diagnose major or primary failures within the system. At a minimum, the diagnostic capability shall include a built-in test (BIT) that verifies basic communication and functionality

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of instrumentation, auxiliary equipment, and major subassemblies that are removable and replaceable at the operator level. The diagnostic capability shall be menu-driven, have a drop-down option, and have an operator-initiated utility of the AGTS application software.

Any diagnostic software with additional troubleshooting capability beyond internal BIT shall be compatible for use on Microsoft Windows 11 operating system computer. This applies to all diagnostic software for any Programmable Logic Controller (PLC), Supervisory Control and Data Acquisition (SCADA), and other control and functional subsystems as well as any engine diagnostic capability (via CAN-bus connector). Any interface hardware/cable (RS-232, USB, CAN-bus) required to communicate between a host computer with diagnostic software and the AGTS shall be provided with each AGTS.

3.10.3 Utility software. Any utility software used for loading/updating of firmware in a PLC, SCADA, or any other logic devices with memory shall be compatible for use on Microsoft Windows 11 operating system computer. Any interface hardware/cable (RS-232, USB, CAN-buss) required to communicate between a host computer with utility software and the AGTS shall be provided with each AGTS.

3.10.4 Software license and activation. Hardware type software keys or dongles shall not be required for software/firmware installation, functionality, or license verification. Additionally, no external internet connection shall be required for software/firmware installation, functionality, or license verification.

3.10.5 Software configuration. All delivered software and firmware shall be of the same version/build during the duration of the contract unless approved by the government.

3.11 Safety.

3.11.1 System safety. The design of the AGTS shall not contain any system safety mishap risk categories greater than medium as defined in MIL-STD-882.

3.11.2 Hazardous materials. The design shall minimize, and control hazards associated with the inclusion or use of hazardous or toxic materials and the generation of toxic or noxious gases. The AGTS shall not generate or use Class I or Class II Ozone Depleting Substances (ODS) during operation, maintenance, or disposal. Class I ODS and hazardous materials shall not be used in any system, component, or process. The AGTS shall not contain or use either hexavalent chromium or cadmium without written approval by the procuring activity. The Environmental Protection Agency maintains an online list of toxic chemicals and hazardous substances at http://www.epa.gov/emergencies/tools.htm. Class I and Class II ODS are defined in 40CFR82.

3.11.3 Component protection. The AGTS shall be designed so that all electrical/mechanical output components are enclosed, and access is not required during normal operation. Further, the AGTS shall be designed to safely permit access to internal components that may be required during troubleshooting for failures or other special needs. All space in which work is performed during operation, service, and maintenance shall be free of hazardous

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protrusions, sharp edges, or other features which may cause injury to personnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, that are of such nature or so located as to be hazardous to personnel, shall be guarded or insulated to eliminate the hazard without impairing the function of the parts. All wires, cables, tubes, and hoses shall be supported and protected to minimize chafing and abrasion and shall be located to provide acceptable clearance from moving parts and high operational temperatures. Grommets shall be provided wherever wires, cables, tubes, or hoses pass through bulkheads, partitions, or structural members. The AGTS shall have, as a minimum the safety protections listed below:

a. Input power disconnect
b. Circuit breakers
c. Emergency stop functionality
d. Shunt trip release
e. Door interlock switches
f. Controlled drive shut down
g. Drive coast shut down
h. Generator field overcurrent
i. Cooling air loss
j. Overvoltage
k. Over frequency
l. Vibration
m. Oil flow or return/case or charge
n. Oil differential pressure
o. Pressure/bulb temperature
p. Motor overtemperature
q. Protective shutdown

3.12 Protective coatings, finish, and corrosion control.

3.12.1 Protective coatings. Materials that deteriorate when exposed to operational conditions normally encountered during the service life of the AGTS shall not be used or shall have means of protection against such deterioration that does not prevent compliance with the performance requirements specified herein. The AGTS shall be constructed of corrosion resistant materials to the maximum extent possible. Surface treatments, pre-treatments and coatings shall be applied to all components that are not corrosion resistant. Surface treatments include electroplated coatings, thermal sprayed coatings, and hard anodize coatings. Surface pre-treatments include thin film anodize, conversion coatings, zinc phosphate, and other coatings used to enhance subsequent corrosion protective coating adhesion and performance. Corrosion protective coatings include corrosion inhibited epoxy primers and polyurethane topcoats, and zinc-rich epoxy powder coating primers with powder coating topcoats, or similar powder coating systems (upon approval of the procuring activity). Corrosion protective coatings that chip, crack, or scale with age or extremes of climatic conditions or when exposed to heat shall not be used. Exposed surfaces of fasteners, handles, and fittings shall also be primed and painted. Commercial items (see 6.4.1) used as components of the AGTS (but not the AGTS itself) may be

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prepared and coated in accordance with the manufacturer's standard practice, provided it is compatible with the exterior finish color. Some commercial items (see 6.4.1) shall have the protective coatings required herein in accordance with the respective material substrates.

3.12.2 Surface preparation. Components shall be cleaned, degreased, and prepared prior to any scuffing, sanding, blasting, or application of any pretreatment. Surface preparation and pretreatment shall be in accordance with the respective primer and topcoat specifications. All corrosion protection steps, regardless of material, shall be applied before any new oxidation or rusting occurs post-treatment and shall be completed prior to application of any respective liquid or powder coating used as either primer or topcoat. Surfaces shall be scuffed, sanded, and blasted, depending upon material type and thickness, to remove any existing surface impurities such as: primer/paint/topcoat, corrosion, and welding/grinding/cutting slag that may prevent adhesion and protection of further corrosion control measure. Any surface undergoing a media blasting process shall have the blasting pressure, blasting material type, and blasting material grit suitably chosen to prevent warpage or other structural damage. Coatings requiring a 'near-white' ferrous metal finish shall have the surface(s) prepared in accordance with SSPC-SP 10/NACE No. 2. General surface finish and cleanliness shall be in accordance with SSPC-Vis 1.

3.12.2.1 Aluminum pretreatment. Aluminum surfaces used in construction of the AGTS shall be treated with an applicable conversion coating prior to application of any other corrosion prevention coating. Anodizing or a MIL-DTL-81706 conversion coating process may be utilized, at the discretion of the AGTS manufacturer. Any anodizing shall be applied in accordance with MIL-A-8625, Type II, Class 1; no further conversion coating is required on anodized aluminum surfaces. Alternatively, aluminum surfaces shall have MIL-DTL-81706, Type II, Class 1A, Form I, Method C chemical conversion coating applied in accordance with MIL-DTL-5541, Type II, Class 1A.

3.12.2.2 Ferrous pretreatment. Ferrous surfaces used in construction of the AGTS shall be treated with a metal etching compound prior to application of any further pretreatment process(es). Ferrous surfaces with iron or zinc phosphate coatings/etches do not require further pretreatment. Ferrous surfaces without iron or zinc phosphate coatings/etches shall be treated with a wash primer in accordance with TT-C-490 to seal out flash rust.

3.12.3 Primer. Raw metal edges, to include fastener and drain holes, shall be coated with primer prior to applying the topcoat. Any surfaces where primer and topcoat are not required, or intended to be applied, shall be masked off using materials and processes acceptable for use with applied primer and topcoat system. A primer color shall be chosen that is compatible with the color of the topcoat used such that scratches in the topcoat do not yield an obtrusive contrast with any exposed primer (tone-down primer in a similar shade/color family as the topcoat).

3.12.3.1 Ferrous surfaces. Ferrous structures and surfaces shall be primed with a non-water reducible zinc rich primer in accordance with MIL-PRF-32550, Type I, Form A, Class S. The zinc rich primer shall be followed, within four hours, by a coat of MIL-DTL-53030 intermediate primer in a wet-to-wet primer application. The primer system shall yield a dry-film thickness (dft) of 2.0-2.5 mils for the zinc rich primer and 0.9 to 1.1 mils for the intermediate

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primer. The primer system shall be allowed to dry and fully cure in accordance with the primer manufacturer's directions prior to applying the topcoat.

3.12.3.2 Aluminum and mixed aluminum and ferrous surfaces. Aluminum and mixed aluminum and ferrous structures and surfaces shall be primed with an epoxy primer in accordance with MIL-PRF-23377, Type I, Class N. This single part primer system shall yield a dft of 0.6 to 0.8 mils.

3.12.4 Topcoat. All AGTS components, except for commercial items (see 6.4.1), plastic, and rubber shall be top coated with polyurethane in accordance with MIL-PRF-85285, Type IV, Class H. Upon request of the contractor, and approval of the procuring activity, MIL-PRF-85285 Type II or Type III coatings may be applied in lieu of MIL-PRF-85285 Type IV coatings in areas and locals where regulatory authority prohibit the use of 420g/l VOC. Neither Chemical Agent Resistant Coating (CARC) nor powder coating shall be used. Topcoat shall be applied to a dft of 1.6 to 2.4 mils in all instances, regardless of the primer system utilized. The coating shall be free from runs, sags, orange peel, or other defects. The contractor shall utilize the topcoat material manufacturer's specifications to accomplish application of the coating system in accordance with the manufacturer's directions to the minimum dft required herein. If a conflict arises between the minimum dft required herein, and the topcoat material manufacturer's specification, the topcoat material manufacturer's specification shall take precedence.

3.12.5 Finish. The exterior finish color of the ATGS and the inner surfaces of compartments shall be the standard color used by the manufacturer.

3.12.6 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall not be in contact with each other. Metal plating or metal spraying of dissimilar base metals to provide electromotively compatible abutting surfaces is acceptable. The use of dissimilar metals when separated by an acceptable insulating material is permitted, except in systems where bridging of insulation materials by an electrically conductive fluid can occur. Sealants or gel type gasket materials shall be used between faying surfaces and butt joints.

3.13 Markings. All external devices which require an operational or maintenance interface (that is, cautions and warnings) shall be marked in accordance with MIL-STD-130. Markings shall be applied with decals and shall be 1-inch-high block letters unless prohibited by the available space. In such cases, the markings shall be the largest size possible but shall not be less than one-half inch high. Markings, Information/Caution shall be Lusterless Black, Color Number 37038 of SAE AMS-STD-595, and Markings, Warning/Danger shall be Lusterless Red, Color Number 31136 of SAE AMS-STD-595.

3.14 Identification and information plates.

3.14.1 Identification plate. The AGTS shall be provided with a permanently marked identification data plate constructed of a non-corroding metal permanently affixed to the exterior of the AGTS in a readily accessible location in accordance with MIL-STD-130. The identification plate shall contain the following information: nomenclature, part number, serial number, date of manufacture, manufacturer's name, Commercial and Government Entity

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(CAGE), date of warranty expiration, contract number, and National Stock Number (NSN). In addition, the identification plate shall include six (6) blank lines across the bottom for future etching by the USAF. The identification plate shall be mounted at a conspicuous place on the exterior of the AGTS.

3.14.2 Unique Identification (UID). Unique Identification (UID) The AGTS and any of its components for which the Government's unit cost is more than $5,000, is serially managed, or the procuring agency determines is mission essential, shall have Unique Identification (UID) (also known as Item Unique Identification (IUID)) information permanently affixed on or near the respective identification plate(s), marked in accordance with MIL-STD-130. UID information shall be included as both a bar code and human readable markings and contain the following information:

a. National Stock Number (NSN).

b. Manufacturer's part number.

c. Serial number.

d. Manufacturer's CAGE.

3.15 Design and construction. The design shall promote cost effective, life-cycle sustainability by addressing considerations such as incorporating open standards while satisfying system performance requirements. It shall be designed and constructed so that no parts will work loose in service, and to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. It shall be weatherproof and designed to prevent the intrusion of water, sand, and dust into critical operating components.

3.15.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials shall be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.15.2 Mechanical design.

3.15.2.1 Fastening devices. All screws, bolts, nuts, pins, and other fastening devices shall be properly designed, manufactured, and installed with the means of preventing loss of torque or adjustment. Cotter pins, lock washers, or nylon patches shall not be used for this purpose, except for the attachment of trim items or as provided in commercial (see 6.4.1) components. When applicable, fasteners shall be installed with the application of a design fastening torque such that an acceptable preload is introduced in the fastener to prevent losing or fatiguing the fasteners. Externally threaded fasteners for use with internally threaded nuts or holes shall be in accordance with SAE J429. Internally threaded nuts shall be in accordance with SAE J995. Tapped threads shall have a minimum thread engagement in accordance with Table III. To the maximum extent possible, fastening designs that utilize bolted connections shall be accomplished using a through-bolt method, and secured with locking nuts or weld-nuts. The use

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of riv-nuts and tapped threads is discouraged. The use of socket-head, flat, countersunk fasteners shall be minimized. When used, socket-head, flat, countersunk fasteners shall be in accordance with NASM 24667, or ASME B18.3 and ASTM F835, and shall be used with a corresponding nut in lieu of a threaded hole to alleviate potential fastener/hole misalignment and bending stress on the head. When in contact with aluminum materials, steel fasteners shall be coated with a Zinc/Aluminum Corrosion Protective Coating in accordance with ASTM F1136 Grade 3.

TABLE III. Minimum thread engagement.

Material	Minimum Thread Engagement
Steel	1.0 times the nominal fastener diameter
Cast iron, brass, or bronze	1.5 times the nominal fastener diameter
Aluminum, zinc, or plastic	2.0 times the nominal fastener diameter

3.15.2.2 Welders and welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof, and service loadings. Cast components, structural steel components with an ultimate tensile strength greater than 100,000 psi, and threaded fasteners (except for weld studs and weld nuts) shall not be welded. Heat treated components shall be in the annealed, normalized condition when welded.

3.15.3 Wiring. All wires used in the AGTS shall be multi-stranded copper conductor, flame retardant, moisture, heat, and oil resistant in accordance with NFPA 70. All wires shall be numbered for easy identification. The marking shall be permanently applied to the wire to withstand all specified operating conditions herein throughout the life expectancy of the end item. Marking shall be applied within 3 inches of each end and at intervals not to exceed 12 inches throughout the wire length. Wires having a total length of less than 18 inches shall be marked within 3 inches of both ends and once in the (approximate) middle of the wire. The wire-numbering scheme shall be consistent with that currently being employed by industry standards. Wires energized from the battery (except the battery cables) shall be marked using red characters. All other wires shall be marked using black characters. The use of colored wire coating as a marking scheme is prohibited. Wires, wire runs, and wiring harnesses shall be placed, bundled, and positioned so that access to other components is not inhibited or impeded. Any tying or lacing used to bundle individual wires into harnesses shall be of sufficient strength to hold the wires in position without crimping, slicing, or otherwise damaging wire insulation or strands. Wires, wire runs, and wiring harnesses shall be supported to retain position and provide support from stretching or sagging. Anti-chafe methods shall be employed to prevent damage from rubbing other components of the AGTS as well as abrasive damage due to sand/dust intrusion. Wires, wire runs, and wiring harnesses shall be protected from abrasive damage when entering, exiting, or passing through panels or bulkheads via the use of appropriately sized grommets.

3.15.3.1 Plugs and connectors. Weatherproof, sealed, O-ring connectors shall be used to the maximum extent possible to join wires and wiring harnesses and prevent the intrusion of

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water, sand, dust, condensation, and salt fog into wiring connections. The use of military specification (MS) type plugs and receptacles is also encouraged. The use of terminal strips and ring or forked style crimp on connections for wire terminations shall be minimized. Unused openings in the plugs, receptacles, switches, or other similar wiring and control devices shall be sealed. Individual wires and wire runs within a wiring harness shall be the full length of the harness and not spliced together. The use of crimp-on 'piggyback' style splices used to join one or more wires together is prohibited.

3.15.4 Foreign object debris (FOD). Metal, rubber, and plastic tags, covers, or plugs commonly affixed to commercial items (see 6.4.1) during the manufacturing process shall be removed from the AGTS to prevent FOD hazards. All loose metal parts, such as pins, connector covers, and caps shall be securely attached to the AGTS with wire ropes or chains. "Dog tag" style beaded chains shall not be provided. Removable panels, if provided, shall be attached with captive fasteners.

3.15.5 Error-proof design. The design of the AGTS shall incorporate error-proofing in equipment mounting, installing, interchanging, connecting, and operating.

a. Equipment shall include physical features (for example, supports, guides, size, or shape differences, fastener locations, and alignment pins) that prevent improper mounting. In the absence of physical features, equipment shall be labeled or coded to identify proper mounting and alignment.

b. Equipment that has the same form and function shall be interchangeable throughout a system and related systems. If equipment is not interchangeable functionally, it shall not be interchangeable physically.

c. Connectors serving the same or similar functions shall be designed to preclude mis-mating or misalignment.

d. Design, location, procedural guidance, and applicable warning labels shall be provided to prevent damage to equipment while it is being handled, installed, operated, or maintained.

3.16 Workmanship. The AGTS, including all parts and accessories, shall be constructed, and finished in a thoroughly workmanlike manner. Workmanship objectives shall include freedom from blemishes, defects, burrs and sharp corners and edges; accuracy of dimensions, surface finish, and radii of fillets; marking of parts and assemblies; and proper alignment of parts and tightness of assembly fasteners.

3.16.1 Bolted connections. Bolt holes shall be accurately punched or drilled and shall be deburred. Threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.16.2 Riveted connections. Rivet holes shall be accurately punched or drilled and shall be deburred. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet

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heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the component.

3.16.3 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustment, gear assemblies shall be free of excessive backlash.

3.16.4 Cleaning. The AGTS shall be thoroughly cleaned. Loose, spattered, or excess solder; welding slag; stray bolts, nuts, and washers; rust; metal particles; pipe compound; and other foreign matter shall be removed during and after final assembly.

3.16.5 Touch-up. Any minor dings, scuffs, or scratches that may occur during the assembly process shall be cleaned, scuffed (as required), primed (thin tie coat), and painted in accordance with the corrosion prevention requirements stated herein, using the same products as applied to the surface(s) prior to assembly. Surfaces that were originally powder coated shall utilize liquid-based material(s) such as MIL-DTL-53022 or MIL-PRF-23377 primers and MIL-PRF-85285 matching color shade topcoat for touch-up.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Preproduction inspection (see 4.2).

b. Operational inspection (see 4.3).

c. Conformance inspection (see 4.4).

Requirements shall be verified in accordance with Table IV.

Table IV. Requirement verification matrix.

Section 3 Requirement	Verification Method	Section 4 Verification
3.1 Preproduction.	Examination	4.6.1 Examination of product.
3.2 Description and general layout.	Examination	4.6.1 Examination of product.
3.3 Performance.	Test	4.6.2 Performance tests.
3.3.1 AGTS system.	Test	4.6.2 Performance tests.
3.3.1.1 Mechanical Power/ Motor System.	Test	4.6.2 Performance tests.
3.3.1.1.1 Hydraulic supply.	Test	4.6.2 Performance tests.
3.3.1.1.2 Heat exchanger.	Test	4.6.2 Performance tests.
3.3.1.1.3 Supply pump.	Test	4.6.2 Performance tests.
3.3.1.1.4 Scavenge pump.	Test	4.6.2 Performance tests.
3.3.1.1.5 Reservoir.	Test	4.6.2 Performance tests.

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Table IV. Requirement verification matrix – Continued

Section 3 Requirement	Verification Method	Section 4 Verification
3.3.1.1.6 Filter.	Test	4.6.2 Performance tests.
3.3.1.1.7 Heater.	Test	4.6.2 Performance tests.
3.3.1.1.8 Power Supply/Drive Controller.	Test	4.6.2 Performance tests.
3.3.1.1.9 Resistive/Reactive Load Bank.	Test	4.6.2 Performance tests.
3.3.1.1.10 Flushing.	Test	4.6.2 Performance tests.
3.3.2 Control system.	Test	4.6.2 Performance tests.
3.4 Environmental conditions.		
3.4.1 Operating temperature range.	Test	4.6.3.1 Low temperature storage and operation test.
3.4.1 Operating temperature range.	Test	4.6.3.2 High temperature storage and operation test.
3.4.2 Storage temperature range.	Test	4.6.3.1 Low temperature storage and operation test.
3.4.2 Storage temperature range.	Test	4.6.3.2 High temperature storage and operation test.
3.5 Weight and dimensions.	Test	4.6.4.1 Weight test.
3.5 Weight and dimensions.	Test	4.6.4.2 Dimension measurement.
3.6 Transportability.	Analysis	4.6.5 Transportability verification.
3.7 Reliability, maintainability, and service life.		
3.7.1 Reliability.	Analysis	4.6.6.1 Reliability model and prediction.
	Test	4.6.6.2 Reliability tests. 4.6.6.2.1 Reliability Qualification Test (RQT).
3.7.2 Maintainability.	Demonstration	4.6.6.3 Preventive maintenance demonstration.
3.7.2.1 Preventive maintenance.	Demonstration	4.6.6.3 Preventive maintenance demonstration.
3.7.3 Service Life.	Certification	4.6.1 Examination of product.
3.7.4 Warranty.	Certification	4.6.1 Examination of product.
3.8 Human systems integration (HSI).	Examination	4.6.1 Examination of product.
3.9 Electromagnetic interference (EMI).	Test	4.6.7 Electromagnetic interference test.
3.10 Cybersecurity and software.	Analysis Demonstration	4.6.8 Cybersecurity and software verification.

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Table IV. Requirement verification matrix – Continued

Section 3 Requirement	Verification Method	Section 4 Verification
3.11 Safety.		
3.11.1 System safety.	Analysis	4.6.9 System safety hazard analysis.
3.11.2 Hazardous materials.	Examination	4.6.1 Examination of product.
3.11.3 Component protection.	Examination	4.6.1 Examination of product.
3.12 Protective coatings, finish, and corrosion control.		
3.12.1 Protective coatings.	Examination	4.6.1 Examination of product.
3.12.2 Surface preparation.	Examination	4.6.1 Examination of product.
3.12.2.1 Aluminum pretreatment.	Examination	4.6.1 Examination of product.
3.12.2.2 Ferrous pretreatment.	Examination	4.6.1 Examination of product.
3.12.3 Primer.	Examination	4.6.1 Examination of product.
3.12.3.1 Ferrous surfaces.	Examination	4.6.1 Examination of product.
3.12.3.2 Aluminum and mixed aluminum and ferrous surfaces.	Examination	4.6.1 Examination of product.
3.12.4 Topcoat.	Examination	4.6.1 Examination of product.
3.12.5 Finish.	Examination	4.6.1 Examination of product.
3.12.6 Dissimilar metals.	Examination	4.6.1 Examination of product.
3.13 Markings.	Examination	4.6.1 Examination of product.
3.14 Identification and information plates.		
3.14.1 Identification plate.	Examination	4.6.1 Examination of product.
3.14.2 Unique Identification (UID).	Examination	4.6.1 Examination of product.
3.15 Design and construction.	Examination	4.6.1 Examination of product.
3.15.1 Recycled, recovered, environmentally preferable, or biobased materials.	Examination	4.6.1 Examination of product.
3.15.2 Mechanical design.		
3.15.2.1 Fastening devices.	Examination	4.6.1 Examination of product.
3.15.2.2 Welders and welding.	Examination	4.6.1 Examination of product.
3.15.3 Wiring.	Examination	4.6.1 Examination of product.
3.15.3.1 Plugs and connectors.	Examination	4.6.1 Examination of product.
3.15.4 FOD.	Examination	4.6.1 Examination of product.
3.15.5 Error-proof design.	Examination	4.6.1 Examination of product.
3.16 Workmanship.	Examination	4.6.1 Examination of product.
3.16.1 Bolted connections.	Examination	4.6.1 Examination of product.
3.16.2 Riveted connections.	Examination	4.6.1 Examination of product.
3.16.3 Gear and lever assemblies.	Examination	4.6.1 Examination of product.
3.16.4 Cleaning.	Examination	4.6.1 Examination of product.
3.16.5 Touch-up.	Examination	4.6.1 Examination of product.

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4.2 Preproduction inspection. The Preproduction AGTS shall be subjected to the analyses, certifications, demonstrations, examinations, and tests described in 4.6.1 through 4.6.9. The contractor shall provide or arrange for all test equipment and facilities, except for the compatibility test described in 4.6.10. Unless otherwise approved by the procuring activity, all Preproduction AGTS shall be in the same configuration at all times, and configuration changes shall not be made during the Preproduction inspection.

4.3 Operational inspection. All operational inspection shall be satisfactorily completed and approved prior to final Preproduction acceptance.

4.4 Conformance inspection. Each production AGTS shall be subjected to the examination described in 4.6.1.

4.5 Inspection requirements.

4.5.1 General inspection requirements. Apparatus used in conjunction with the inspections specified herein shall be laboratory precision type, calibrated at proper intervals to ensure laboratory accuracy.

4.5.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

d. Ambient temperature.

e. Ambient humidity.

4.5.3 Test rejection criteria. Throughout all tests specified herein, the AGTS shall be closely observed for the following conditions, which shall be cause for rejection.

a. Failure to conform to design or performance requirements specified herein or in the contractor's technical proposal.

b. Thermal or structural failure of any component, including permanent deformation, or evidence of impending failure.

c. Evidence of excessive wear. If excessive wear is suspected, the original equipment manufacturers (OEM's) specifications or tolerances shall be utilized for making a determination.

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d. Evidence of corrosion or deterioration.

e. Misalignment of components.

f. Conditions that present a safety hazard to personnel during operation, servicing, or maintenance.

g. Interference between the AGTS components.

4.5.4 Failure reporting, analysis, and corrective action system (FRACAS). A closed loop system shall be used to collect data, analyze, and record timely corrective action for all potential failures that occur during the Preproduction and operational tests. The contractor's existing FRACAS shall be utilized with the minimum changes necessary to conform to this specification. The system shall cover all test samples, interfaces between test samples, test instrumentation, test facilities, test procedures, test personnel, and the handling and operating instructions. The contractor shall establish and maintain a FRACAS database; all information shall be entered into the database. Authorized procuring activity personnel shall have read-only access to the FRACAS database.

4.5.4.1 Problem and failure action. At the occurrence of a problem or potential failure that affects satisfactory operation of a test sample, entries shall be made in the appropriate data logs and the failed test sample shall be removed from test, with minimum interruption to the other test samples continuing test.

4.5.4.1.1 Problem and failure reporting. A failure report shall be initiated at the occurrence of each problem or potential failure of the contractor hardware or software, or Government furnished equipment (GFE). The report shall contain the information required to permit determination of the origin and correction of the failure. The contractor's existing failure report form may be used with minimum changes necessary to conform to the requirements of this specification. The failure report form shall include the information specified in a through c:

a. Descriptions of failure symptoms, conditions surrounding the failure, failed hardware identification, and operating time (or cycles) at the time of failure.

b. Information on each independent and dependent failure and the extent of confirmation of the failure symptoms, the identification of failure modes, and a description of all repair actions taken to return the test sample to operational readiness.

c. Information describing the results of the investigation, the analysis of all part failures, an analysis of the system design, and the corrective action taken to prevent failure recurrence. If no corrective action is taken, the rationale for this decision shall be recorded.

4.5.4.1.2 Identification and control of failed items. A failure tag shall be affixed to the failed part immediately upon the detection of any failure or suspected failure. The failure tag shall provide space for the failure report serial number and for other pertinent entries from the

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test sample failure record. All failed parts shall be marked conspicuously or tagged and controlled to ensure disposal in accordance with contract requirements. Failed parts shall not be handled in any manner which may obliterate facts which might be pertinent to the analysis. Failed parts shall be stored pending disposition by the authorized approval agency of the failure analysis.

4.5.4.1.3 Problem and failure investigations. An investigation and analysis of each reported failure shall be performed. Investigation and analysis shall be conducted to the level of hardware or software necessary to identify causes, mechanisms, and potential effects of the failure. Any applicable method (for example, test, microscopic analysis, applications study, dissection, X-ray analysis, spectrographic analysis) of investigation and analysis which may be needed to determine failure cause shall be used. When the removed part is not defective or the cause of failure is external to the part, the analysis shall be extended to include the circuit, higher hardware assembly, test procedures, and subsystem if necessary. Investigation and analysis of GFE failures shall be limited to verifying that the GFE failure was not the result of the contractor's hardware, software, or procedures. This determination shall be documented for notification of the procuring activity.

4.5.4.1.4 Failure verification. Reported failures shall be verified as actual failures or an acceptable explanation provided to the procuring activity for lack of failure verification. Failure verification is determined either by repeating the failure mode of the reported part or by physical or electrical evidence of failure (for example, leakage residue or damaged hardware). Lack of failure verification, by itself, is not sufficient rationale to conclude the absence of a failure.

4.5.4.1.5 Corrective action. When the cause of failure has been determined, a corrective action shall be developed to eliminate or reduce the recurrence of the failure. Repairs shall be made in accordance with normal field operating procedures and manuals. The procuring activity shall review the corrective actions at the scheduled test status review prior to implementation. In all cases the failure analysis and the resulting corrective actions shall be documented.

4.5.4.1.6 Problem and failure tracking and closeout. The closed loop failure reporting system shall include provisions for tracking problems, failures, analyses, and corrective actions. Status of corrective actions for all problems and failures shall be reviewed at scheduled test status reviews. Problem and failure closeout shall be reviewed to assure their adequacy.

4.6 Test methods.

4.6.1 Examination of product. Each AGTS shall be examined to verify compliance with the requirements herein prior to accomplishing any other demonstrations or tests listed in 4.6. A contractor-generated, Government-approved checklist (part of the test procedure) shall be used to identify each requirement not verified by an analysis, certification, demonstration, or test, and shall be used to document the examination results. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and sealants and their application, fastening, and markings. Proper operation of each AGTS function shall be verified. Each production AGTS shall be inspected to a Government-approved reduced version of the checklist.

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4.6.2 Performance tests.

4.6.2.1 Speed range. The AGTS shall be connected to 220 or 440 volts, 3 phase, 60 cycle power source and operated over the entire speed range. The minimum speed shall not exceed 2440 rpm. The maximum speed shall be 9150 rpm. The test stand shall operate smoothly at any chosen output speed. (See Figures 1 and 2) Failure to meet this speed range shall be cause for rejection.

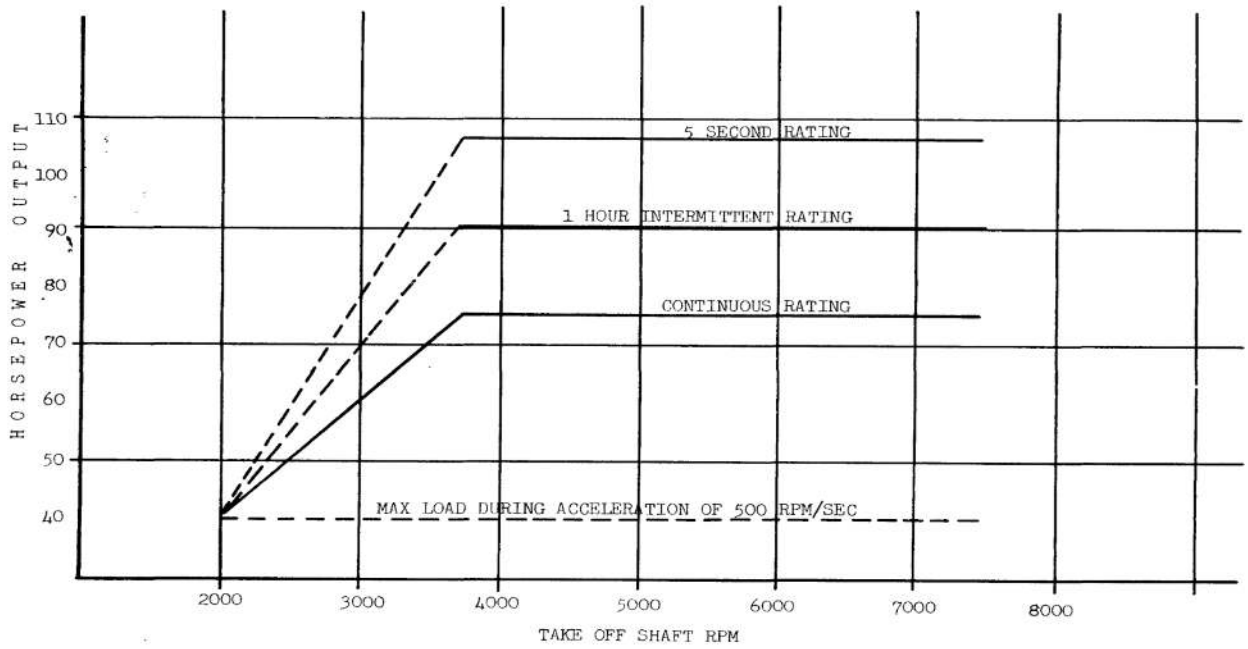


FIGURE 1. Power Output Curve (Low Speed Head Range 2000-7500 RPM).

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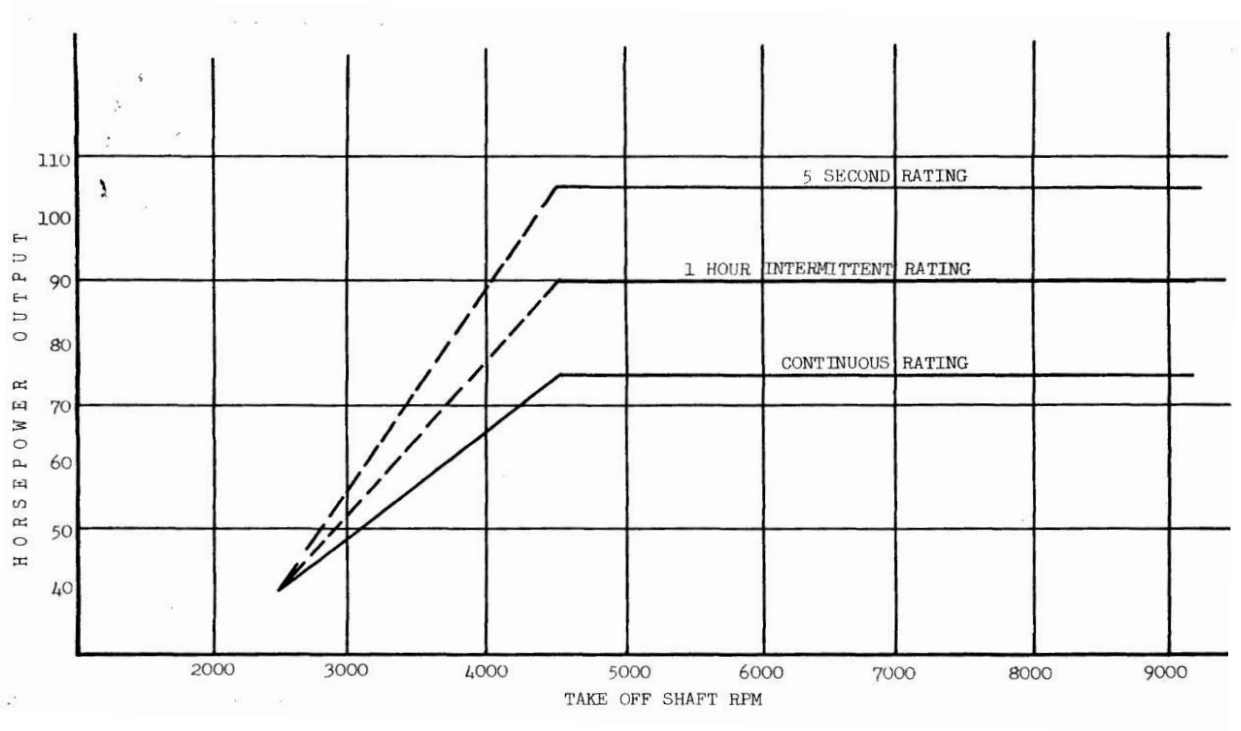


FIGURE 2. Power Output Curve (High Speed Head Range 2440-9150 RPM).

4.6.2.2 Field test performance. The AGTS, modified by an adapter set for a specific constant speed drive application, together with the specific drive components, generator, voltage regulator, and associated control panel mounted on the AGTS, shall be tested to determine its ability to perform the following field tests in accordance with the applicable content speed drive specification:

a. Single unit tests

1) No load

2) Rated load

3) Speed regulation

4) Slip check

5) Pressure switch tests (as applicable)

6) Parallel tests (as applicable)

4.6.2.3 Insulation resistance test. All commercial electrical components shall be tested. The AGTS shall withstand an AC voltage equal to twice the normal operating voltage plus 1000 or 1500 volts rms, whichever is greater. This shall be accomplished by disconnecting the ground

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lead on AGTS, connecting all terminals, and applying the voltage between these terminals and the frame of the stand.

4.6.2.4 Meter calibration. The accuracy of all meter components shall be determined by comparison with standard laboratory type instruments. The accuracy shall be as specified in MIL-DTL-16034B. Accuracy of meters not covered by MIL-DTL-16034B shall be within 3 percent or better.

4.6.2.5 Heat run. The alternator loading the constant speed drive shall be loaded to 17 ±2 kw with a drive input speed of 3000 rpm. The load shall be maintained for 3 hours. During this period, all temperatures shall remain at a safe level. The drive oil in temperature shall be varied from 140 to 250 °F during this test. The test shall be repeated, except the drive input speed shall be 7000 rpm at 30 ±2 kw.

4.6.2.6 Constant speed drive, generator, and panel tests. The AGTS modified by an adapter set for a specific constant speed drive application, together with the specific drive components, generator and regulator and associated control panel mounted on the stand, shall be tested to determine its ability and load of the test stand as specified herein.

a. The following tests, as applicable, shall be conducted in accordance with applicable constant speed drive specification:

    1) No load

    2) Rated load (to power limits of prime mover)

    3) Speed regulation

    4) Parallel operation

    5) Load division

    6) Governor overspeed circuit

    7) Under speed switch

    8) Sudden load application

    9) Overloads (to the power output limits of the prime mover)

b. The following generator tests shall be conducted in accordance with MIL-PRF-21480B:

    1) Maximum speed for regulation

    2) Voltage regulation (at room temperature)

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4.6.3 Environmental testing.

4.6.3.1 Low temperature storage and operation test. A Preproduction AGTS shall be tested in accordance with MIL-STD-810, Method 502.7, Procedures I and II, to demonstrate compliance with the low temperature storage and operating requirements of 3.4.1 and 3.4.2. Test duration shall be one 24-hour cycle for each procedure beginning no less than two hours after test item temperature stabilization.

4.6.3.2 High temperature storage and operation test. A Preproduction AGTS shall be tested in accordance with MIL-STD-810, Method 501.7, Procedures I and II, to demonstrate compliance with the high temperature storage and operating requirements of 3.4.1 and 3.4.2. Test duration shall be one 24-hour cycle for each procedure beginning no less than two hours after test item temperature stabilization.

4.6.4 Weight and dimension tests.

4.6.4.1 Weight test. The weight of a Preproduction AGTS shall be measured to demonstrate compliance with the weight requirement of 3.5.

4.6.4.2 Dimension measurement. A Preproduction AGTS shall be measured to demonstrate compliance with the dimensional requirements of 3.5.

4.6.5 Transportability verification. An engineering analysis shall be performed to demonstrate compliance with 3.6.

4.6.6 Reliability and maintainability – analysis, demonstration, and tests.

4.6.6.1 Reliability model and prediction. The contractor shall develop and maintain a basic reliability model for the system in accordance with Appendix A, A.4.1. The contractor shall prepare and maintain a basic reliability prediction for the system in accordance with A.4.2; it shall be based upon the associated basic reliability model.

4.6.6.2 Reliability tests. All the preproduction AGTS shall be subjected to a Reliability Qualification Test (RQT) in accordance with 4.6.6.1. All the AGTS shall be operated continuously throughout the entire test period, except as approved by the procuring activity or when an AGTS is taken offline for a PMI or investigation or repair of a failure or implementation of a corrective action (see A.3.1). The time accumulated while an AGTS is offline shall not be included in the test time. PMIs shall be staggered, if possible, so that only one AGTS is offline for a PMI at any time. All requirements of Appendix A shall apply to the reliability tests. Failure definitions shall be in accordance with A.3.3—A.3.8.

4.6.6.2.1 RQT. A 50-hour fixed duration RQT shall be performed to demonstrate compliance with 3.7. Nominal consumer's and producer's risks shall be 20 percent; the discrimination ratio shall be 3, and no more than 2 failures shall be allowed. (Ref. Test Plan XVII-D of MIL-HDBK-781.) Failure suppression, the removal of a failure from the RQT based

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upon the assumption that a future corrective action will eliminate that failure mode, shall not be allowed.

4.6.6.3 Preventive maintenance demonstration. All recommended preventive maintenance tasks shall be performed, and the task times shall be recorded. It shall be demonstrated that the forces required do not exceed those allowed in MIL-STD-1472.

4.6.7 Electromagnetic interference test. A Preproduction AGTS shall be tested in accordance with MIL-STD-461: RE 102 and RS 103 to demonstrate compliance with 3.9.

4.6.8 Cybersecurity and software verification. A vulnerability and security analysis shall be conducted on the AGTS and associated software along with demonstration of software functionality and implemented technical security controls to demonstrate compliance with cybersecurity and software requirements of 3.10.

4.6.9 System safety hazard analysis. A system safety hazard analysis of the AGTS shall be conducted in accordance with 4.3.1 through 4.3.8 of MIL-STD-882 to demonstrate compliance with the mishap risk requirement of 3.11.1.

4.6.10 Compatibility test. Compatibility testing for one preproduction AGTS shall be coordinated and performed at Seymour Johnson AFB, North Carolina.

5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. The AGTS covered by this specification is intended for use in testing aircraft alternating current electrical power systems.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. If Preproduction inspection is required (see 3.1).

c. Packaging requirements (see 5.1).

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6.3 Subject term (key word) listing.

A-10
AGTS
B-1
F-15C
F-15E
F-15EX
F-16
Hydraulic supply
K400B
K738XL/K400B-XL
MC-2000
Power supply/drive controller and console
Resistive/reactive load bank

6.4 Definitions.

6.4.1 Commercial item. (1) Any item, other than real property, that is of a type customarily used by the general public or by non-Governmental entities for purposes other than Governmental purposes, and—

(i) Has been sold, leased, or licensed to the general public; or

(ii) Has been offered for sale, lease, or license to the general public;

(2) Any item that evolved from an item described in paragraph (1) of this definition through advances in technology or performance and that is not yet available in the commercial marketplace, but will be available in the commercial marketplace in time to satisfy the delivery requirements under a Government solicitation;

(3) Any item that would satisfy a criterion expressed in paragraphs (1) or (2) of this definition, but for—

(i) Modifications of a type customarily available in the commercial marketplace; or

(ii) Minor modifications of a type not customarily available in the commercial marketplace made to meet Federal Government requirements. Minor modifications means modifications that do not significantly alter the non-Governmental function or essential physical characteristics of an item or component, or change the purpose of a process. Factors to be considered in determining whether a modification is minor include the value and size of the modification and the comparative value and size of the final product. Dollar values and percentages may be used as guideposts, but are not conclusive evidence that a modification is minor;

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(4) Any combination of items meeting the requirements of paragraphs (1), (2), (3), or (5) of this definition that are of a type customarily combined and sold in combination to the general public;

(5) Installation services, maintenance services, repair services, training services, and other services if—

(i) Such services are procured for support of an item referred to in paragraph (1), (2), (3), or (4) of this definition, regardless of whether such services are provided by the same source or at the same time as the item; and

(ii) The source of such services provides similar services contemporaneously to the general public under terms and conditions similar to those offered to the Federal Government;

(6) Services of a type offered and sold competitively in substantial quantities in the commercial marketplace based on established catalog or market prices for specific tasks performed or specific outcomes to be achieved and under standard commercial terms and conditions. For purposes of these services—

(i) "Catalog price" means a price included in a catalog, price list, schedule, or other form that is regularly maintained by the manufacturer or vendor, is either published or otherwise available for inspection by customers, and states prices at which sales are currently, or were last, made to a significant number of buyers constituting the general public; and

(ii) "Market prices" means current prices that are established in the course of ordinary trade between buyers and sellers free to bargain and that can be substantiated through competition or from sources independent of the offerors.

(7) Any item, combination of items, or service referred to in paragraphs (1) through (6) of this definition, notwithstanding the fact that the item, combination of items, or service is transferred between or among separate divisions, subsidiaries, or affiliates of a contractor; or

(8) A non-developmental item, if the procuring agency determines the item was developed exclusively at private expense and sold in substantial quantities, on a competitive basis, to multiple State and local Government. (Reference the Federal Acquisition Regulation (FAR) 2.101)

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APPENDIX A

RELIABILITY AND MAINTAINABILITY

A.1 SCOPE

A.1.1 Scope. This appendix provides definitions of reliability and maintainability terms for use during the preproduction and operational tests. It provides procedures for performing reliability models and predictions, a Failure Modes, Effects, and Criticality Analysis (FMECA), a Reliability-Centered Maintenance (RCM) analysis, and maintainability predictions. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

A.2.1 Government documents.

A.2.1.1 Specifications, standards, and handbooks. The following standard and handbooks of the exact revision listed below form a part of this specification to the extent specified herein.

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-189	Reliability Growth Management
MIL-HDBK-217	Reliability Prediction of Electronic Equipment
MIL-HDBK-338	Electronic Reliability Design Handbook
MIL-HDBK-470	Designing and Developing Maintainable Products and Systems
MIL-HDBK-781	Handbook for Reliability Test Methods, Plans, and Environments for Engineering, Development, Qualification, and Production

(Copies of these documents are available online at https://assist.dla.mil.)

A.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications of the exact revision level shown form a part of this document to the extent specified herein.

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NAVAL SURFACE WARFARE CENTER CARDEROCK DIVISION (NSWCCD)

NSWC-11	Handbook of Reliability Prediction Procedures for Mechanical Equipment, May 2011

(Copies of this document are available online at: http://www.navsea.navy.mil/nswc/carderock/pub/mechrel/products/handbook.aspx.)

A.2.3 Non-Government publications. The following documents of the exact revision listed below form a part of this document to the extent specified herein.

RELIABILITY INFORMATION ANALYSIS Center (RIAC)

NPRD	Nonelectronic Parts Reliability Data
FMD	Failure Mode/Mechanism Distributions

(Application for copies should be addressed to Reliability Information Analysis Center, 201 Mill Street, Rome NY 13440.)

A.2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

A.3 DEFINITIONS

A.3.1 Corrective action. A fix; that is, a change to the design, operation, and maintenance procedures, or to the manufacturing process of the item for the purpose of improving the reliability. (Reference MIL-HDBK-189)

A.3.2 Discrimination ratio (d). The discrimination ratio (d) is one of the standard reliability qualification tests (RQT) plan parameters; it is the ratio of the upper test MTBF (θ_0) to the lower test MTBF (θ_1) that is, $d = \frac{\Theta_0}{\Theta_1}$. (Reference MIL-HDBK-781)

A.3.3 Failure. The event, or inoperable state, in which any item, or part of an item, does not, or would not, perform as previously specified. (Reference MIL-HDBK-470)

A.3.4 Failure, chargeable. A failure that is not non-chargeable. Replacement of a consumable item (such as a filter or a fan belt) at the preventive maintenance interval (PMI) is not a chargeable failure.

A.3.5 Failure, non-chargeable. A failure that is a non-relevant (see A.3.6); that is induced by Government furnished equipment (GFE) operating, maintenance, or repair procedures; or of a part having a specified life expectancy and operated beyond the specified replacement time of the part. A nonrecurring fluid or recurring fluid leak that does or does not

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result in formation of a droplet shall be non-chargeable unless the leak causes the system to be unsafe. A recurring fluid leak where a droplet forms and falls and any recurring fluid leak where the frequency of droplets makes a measurable stream shall be chargeable. Light bulb and label failures shall be non-chargeable.

A.3.6 Failure, intermittent. Failure for a limited period of time, followed by the item's recovery of its ability to perform within specified limits without any remedial action. (Reference MIL-HDBK-470)

A.3.7 Failure, non-relevant. A failure caused by installation damage; accident or mishandling; failure of the test facility or test-peculiar instrumentation; an externally applied overstress condition, in excess of the approved test requirements; normal operating adjustments specified in the approved operating instructions; or human error. A secondary failure within the test sample, which is directly caused by a non-relevant or relevant primary failure, is also a non-relevant failure. The secondary failure has to be proved to be dependent on the primary failure.

A.3.8 Failure, relevant. An intermittent failure; an unverified failure (a failure which cannot be duplicated, which is still under investigation or for which no cause could be determined); a verified failure not otherwise excluded as a non-relevant failure; or a pattern failure.

A.3.9 Fix. A fix is a corrective action that results in a change to the design, operation, and maintenance procedures, or to the manufacturing process of the item for the purpose of improving the reliability. (Reference MIL-HDBK-189)

A.3.10 Functionally significant item (FSI). An FSI is an item whose failure meets one of the following criteria.

a. Could affect safety.

b. Could result in severe damage or equipment loss.

c. Could degrade or negate mission accomplishment (including survivability provisions).

d. Could have a significant economic impact.

e. Could have one or more of the above effects in combination with another item failure (hidden function).

f. Could have secondary consequences leading to one or more of the above effects.

g. Could have one or more of the above effects due to environmental deterioration or accidental damage.

A.3.11 Mean time between failure (MTBF). Mean time between failure (MTBF) is a basic measure of reliability for repairable items. The mean number of life units during which all

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parts of the item perform within their specified limits, during a particular measurement interval under stated conditions.

A.3.12 MTBF, lower test (θ_1). Lower test MTBF (θ_1) is that value which is the minimum acceptable. The standard RQT plans will reject, with high probability, equipment with a true MTBF that approaches (θ_1). The lower test MTBF is the required MTBF. (Reference MIL-HDBK-781)

A.3.13 MTBF, upper test (θ_0). Upper test MTBF (θ_0) is an acceptable value of MTBF equal to the discrimination ratio (d) times the lower test MTBF (θ_1). The standard RQT plans will accept, with high probability, equipment with a true MTBF that approaches (θ_0). This value (θ_0) should be realistically attainable, based on experience and information. The upper test MTBF is also known as the "design to" MTBF. (Reference MIL-HDBK-781)

A.3.14 MTBF, predicted (θ_p). Predicted MTBF (θ_p) is that value of MTBF determined by reliability prediction methods; it is a function of the equipment design and the use environment. Predicted MTBF (θ_p) should be equal to or greater than the upper test MTBF (θ_0) in value, to ensure, with high probability, that the equipment will be accepted during the RQT. (Reference MIL-HDBK-781)

A.3.15 Mean time to repair (MTTR). Mean time to repair (MTTR) is the sum of corrective maintenance times at any specific level of repair, divided by the total number of failures within an item repaired at that level during a particular interval under stated conditions. (Reference MIL-HDBK-470)

A.3.16 Replaceable item (RI). A replaceable item (RI) is any of those physical entities normally removed and replaced to effect repair at the maintenance level for which the prediction is being made. (Reference MIL-HDBK-470)

A.3.17 Repair. A repair is the refurbishment of a failed part or replacement of a failed part with an identical unit in order to restore the system to be fully mission capable. (Reference MIL-HDBK-189).

A.3.18 Risk, consumer's (β). Consumer's risk (β) is the probability of accepting equipment with a true MTBF equal to the lower test MTBF (θ_1). The probability of accepting equipment with a true MTBF less than the lower test MTBF (θ_1) will be less than the consumer's risk (β). (Reference MIL-HDBK-781)

A.3.19 Risk, producer's (α). Producer's risk (α) is the probability of rejecting equipment which has a true MTBF equal to the upper test MTBF (θ_0). The probability of rejecting equipment with a true MTBF greater than the upper test MTBF (θ_0) will be less than the producer's risk (α). (Reference MIL-HDBK-781)

A.4 REQUIREMENTS

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A.4.1 Basic reliability model. The contractor shall develop and maintain a basic reliability model for the entire AGTS. All equipment and associated quantities comprising these parts shall be included in the model to at least Level 3 of the Bill of Materials (BOM). All equipment, including those intended solely for item redundancy and alternate modes of operation, shall be modeled in series. A basic reliability block diagram shall be developed and maintained for the items with associated allocations and predictions in each reliability block. The basic reliability block diagram shall be keyed and traceable to functional block diagrams, drawings, and schematics, and shall provide the basis for accurate mathematical representation of basic reliability. Nomenclature of elements of the item used in the basic reliability block diagrams shall be consistent with that used in functional block diagrams, drawings, schematics, weight statements, power budgets, and specifications.

A.4.2 Basic reliability prediction. The contractor shall prepare and maintain a basic reliability prediction for the entire AGTS as described in 3.7; it shall be based upon the associated basic reliability model (see A.4.1). All equipment and associated quantities comprising these parts shall be included in the model except for documented exclusions approved by the procuring activity. Failure rate data (or equivalent reliability parameters) shall be consistent with the level of detail of the basic reliability model. The prediction shall be based upon the worst-case service use profile. All data sources for failure rates, failure distribution, and failure rate adjustment factors (for example, stress factors and duty cycle) shall be identified for each reliability block. Data sources shall be MIL-HDBK-217, NPRD (any version or date), NWSC-11, or as otherwise approved by the procuring activity; ground mobile (G_M) data shall be used. Note that these data sources do not contain failure rate data for all possible components in all possible environments and the data for a specific component may not be relevant for any particular application; these data sources shall be supplemented as required.

A.4.3 Failure modes, effects, and criticality analysis (FMECA). The contractor shall perform a FMECA on all FSIs utilizing the basic reliability model (see A.4.1). The body of the FMECA shall be a table with following information provided for each failure mode. In the discussion below, the term “item” refers to the part, subsystem, system, or end item, as appropriate.

A.4.3.1 Identification number. The contractor shall assign an identification number to each failure mode of each item for traceability purposes; it shall identify both the block in the reliability model and the specific failure mode related to that block.

A.4.3.2 Nomenclature. The nomenclature for the item shall be listed; the nomenclature shall be consistent with that used in the basic reliability model.

A.4.3.3 Functions. A concise statement of the functions performed by the hardware item shall be listed. These functions shall include the inherent function(s) of the item, secondary function(s), and relationship(s) to interfacing items.

A.4.3.4 Failure modes and causes. All predictable failure modes for each indenture level analyzed shall be identified and described. Potential failure modes shall be determined by examination of item outputs and functional outputs identified in applicable block diagrams and

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schematics. Failure modes of the individual item function shall be postulated on the basis of the stated requirements in the system definition narrative and the failure definitions included in the ground rules. The most probable causes associated with the postulated failure mode shall be identified and described. Since a failure mode may have more than one cause, all probable independent causes for each failure mode shall be identified and described. The failure causes within the adjacent indenture levels shall be considered. For example, failure causes at the third indenture level shall be considered when conducting a second indenture level analysis. Where functions shown on a block diagram are performed by a replaceable module in the system, a separate FMECA shall be performed on the internal functions of the module, viewing the module as a system. The effects of possible failure modes in the module inputs and outputs describe the failure modes of the module when it is viewed as an item within the system. To assist in assuring that a complete analysis is performed, each failure mode and output function shall, as a minimum, be examined in relation to the following typical failure conditions:

a. Premature operation.

b. Failure to operate at a prescribed time.

c. Intermittent operation.

d. Failure to cease operation at a prescribed time.

e. Loss of output or failure during operation.

f. Degraded output or operational capability.

g. Other unique failure conditions, as applicable, based upon system characteristics and operational requirements or constraints.

FMD (any version or date) shall be used as a data source for failure modes and distributions.

A.4.3.5 Mission phase and operational mode. A concise statement of the mission phase and operational mode in which the failure occurs shall be provided. Where subphase, event, or time can be defined from the system definition and mission profiles, the most definitive timing information should also be entered for the assumed time of failure occurrence.

A.4.3.6 Failure effects. The consequences of each assumed failure mode on item operation, function, or status shall be identified, evaluated, and recorded. Failure effects shall focus on the specific block diagram element which is affected by the failure under consideration. The failure under consideration may impact several indenture levels in addition to the indenture level under analysis; therefore, “local,” “next higher level,” and “end item” effects shall be evaluated. Failure effects shall also consider the mission objectives, maintenance requirements, and personnel and system safety.

A.4.3.6.1 Local effects. Local effects concentrate specifically on the impact an assumed failure mode has on the operation and function of the item in the indenture level under

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consideration. The consequences of each postulated failure affecting the item shall be described along with any second-order effects which result. The purpose of defining local effects is to provide a basis for evaluating compensating provisions and for recommending corrective actions. It is possible for the "local" effect to be the failure mode itself.

A.4.3.6.2 Next higher assembly effects. Next higher-level effects concentrate on the impact an assumed failure has on the operation and function of the items in the next higher indenture level above the indenture level under consideration. The consequences of each postulated failure affecting the next higher indenture level shall be described.

A.4.3.6.3 End item effects. End item effects evaluate and define the total effect an assumed failure has on the operation, function, or status of the uppermost system. The end item effect described may be the result of a double failure. For example, failure of a safety device may result in a catastrophic end effect only in the event that both the prime function goes beyond limit for which the safety device is set, and the safety device fails. Those end effects resulting from a double failure shall be indicated on the worksheets.

A.4.3.7 Failure detection method. A description of the methods by which occurrence of the failure mode is detected by the operator shall be recorded. The failure detection means, such as visual or audible warning devices, automatic sensing devices, sensing instrumentation, other unique indications, or none shall be identified.

A.4.3.7.1 Other indications. Descriptions of indications which are evident to an operator that a system has malfunctioned or failed, other than the identified warning devices, shall be recorded. Proper correlation of a system malfunction or failure may require identification of normal indications as well as abnormal indications. If no indication exists, identify if the undetected failure could jeopardize the mission objectives or personnel safety. If the undetected failure allows the system to remain in a safe state, a second failure situation shall be explored to determine whether or not an indication would be evident to an operator. Indications to the operator shall be described as follows:

a. Normal. An indication that is evident to an operator when the system or equipment is operating normally.

b. Abnormal. An indication that is evident to an operator when the system has malfunctioned or failed.

c. Incorrect. An erroneous indication to an operator due to the malfunction or failure of an indicator (that is, instruments, sensing devices, and visual or audible warning devices).

A.4.3.7.2 Isolation. Describe the most direct procedure that allows an operator to isolate the malfunction or failure. An operator will know only the initial symptoms until further specific action is taken such as performing a more detailed built-in-test (BIT). The failure being considered in the analysis may be of lesser importance or likelihood than another failure that could produce the same symptoms and this shall be considered. Fault isolation procedures require a specific action or series of actions by an operator, followed by a check or cross

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reference either to instruments, control devices, circuit breakers, or combinations thereof. This procedure shall be followed until a satisfactory course of action is determined.

A.4.3.8 Compensating provisions. The compensating provisions, either design provisions or operator actions, which circumvent or mitigate the effect of the failure shall be identified and evaluated. This step is required to record the true behavior of the item in the presence of an internal malfunction or failure.

A.4.3.8.1 Design provisions. Compensating provisions which are features of the design at any indenture level that nullify the effects of a malfunction or failure, control, or deactivate system items to halt generation or propagation of failure effects or activate backup or standby items or systems shall be described. Design compensating provisions include:

a. Redundant items that allow continued and safe operation.

b. Safety or relief devices such as monitoring or alarm provisions which permit effective operation or limits damage.

c. Alternative modes of operation such as backup or standby items or systems.

A.4.3.8.2 Operator actions. Compensating provisions which require operator action to circumvent or mitigate the effect of the postulated failure shall be described. The compensating provision that best satisfies the indication(s) observed by an operator when the failure occurs shall be determined. This may require the investigation of an interface system to determine the most correct operator action(s). The consequences of any probable incorrect action(s) by the operator in response to an abnormal indication should be considered and the effects recorded.

A.4.3.9 Risk priority number (RPN). The contractor shall perform a Criticality Analysis (CA) by calculating a Risk Priority Number (RPN) for each potential failure mode. The RPN shall be calculated by:

$$RPN = S \times O \times D$$

where:

S = Severity (see A.4.3.9.1),
O = Occurrence (see A.4.3.9.2), and
D = Detection (see A.4.3.9.3).

A.4.3.9.1 Severity (S). Severity (S) is an assessment of the effect of the potential failure mode. It shall be estimated on a scale of “1” to “10” using Table 7.8-3 of MIL-HDBK-338. Severity applies only to the effect of the failure.

A.4.3.9.2 Occurrence (O). Occurrence (O) is the likelihood that a specific failure mode will occur. The likelihood of occurrence ranking number is an index number, rather than a probability, although it is based upon a probability. Removing or controlling one of more of the

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causes or mechanisms of the failure mode, through a design change, is the only way a reduction in occurrence ranking can be obtained. The likelihood of occurrence ranking number shall be estimated on a scale of "1" to "10" using Table 7.8-4 of MIL-HDBK-338.

A.4.3.9.3 Detection (D). Detection (D) is an assessment of the ability of the design control to detect a potential cause or mechanism or to detect the failure mode. In order to achieve a lower detection ranking, the planned design control has to be improved. The detection ranking number shall be estimated on a scale of "1" to "10" using Table 7.8-5 of MIL-HDBK-338.

A.4.4 Reliability-centered maintenance (RCM).

A.4.4.1 Purpose. The purpose of RCM is to identify both field and depot level preventive maintenance requirements.

A.4.4.2 Analysis method. The decision logic described below shall be applied to each significant item (system, subsystem, module, component, accessory, unit, and part) using the technical data available. Principally, the evaluations are based on the item's functional failures and failure causes. The method is to:

a. Identify the FSIs. Select the FSIs beginning at the system level descending as necessary through the part level. All items whose known or anticipated failure could adversely affect safety and mission success or have significant economic effects shall be included in the list of FSIs.

b. Identify "Other Items". During the identification process for FSIs, some items may be recognized as not meeting the criteria for an FSI, but experience with like or similar items and recommendations for unique items may indicate a maintenance task is desirable. These nonsignificant items are categorized as "other items" and are dispositioned separately in the decision logic.

c. Accomplish a FMECA on each FSI in accordance with A.4.3. All predictable and potential failure modes and causes shall be fully considered in the FMECA, including those for environmental deterioration and accidental damage.

d. Utilizing the RCM decision logic, identify the tasks to be accomplished. If no applicable and effective task can be identified, redesign of equipment is required in the safety effects categories, redesign may be required in the mission effects categories, and redesign may be desirable in the nonmission economic effects categories.

    (1) Failures with direct adverse mission affects that cannot be prevented through preventive maintenance compromise the ability of the equipment to accomplish its intended functions. Since equipment may have multiple capabilities and a failure may not affect all of them or affect them all to the same degree, redesign shall be considered within the total set of equipment capabilities and mission scenarios. Redesign is required to allow full mission capability. If the equipment after the failure has partial mission capabilities, redesign shall be weighed against

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the degree and criticality of the mission impact to determine if the use of program resources for redesign is warranted.

(2) Failures with nonmission economic effects that are not preventable through preventive maintenance will not compromise safety or mission; however, the economic penalties to allow items to fail may be so severe that it would be more advantageous to redesign.

e. Identify the intervals for task development.

f. Identify the recommended level of maintenance for accomplishing the tasks.

A.4.4.2.1 Task analysis. Prior to applying the RCM decision logic to an item, the FSI, its function, functional failure, failure mode, failure effect, and any additional data pertinent to the item, such as the manufacturer's part number, a brief description of the item, expected failure rate, hidden functions, and redundancies, shall be documented. Each functional failure and failure mode shall be processed through the logic so that a judgment can be made as to the necessity of a task. The resultant tasks and intervals shall be included in the preventive maintenance program.

A.4.4.2.2 Decision logic. The decision logic shall be used for analysis of systems and equipment items. The decision logic has three levels.

a. The first level (questions 1 and 2) categorizes the FSIs and “other items.” The logic for “other items” leads to appropriate tasks and intervals based on experience or manufacturers' recommendations. No further analysis is required for "other items.” The logic for FSIs leads to the second and third level questions.

b. The second level (questions 3, 4, 5, 6, and 7) requires evaluation of each functional failure for determination of the ultimate effect category, that is, evident safety, mission, or nonmission economic effects, or hidden safety, mission, or nonmission economic effects.

c. The third level (questions A through F or A through E, as applicable) then takes the failure modes for each functional failure into account for selecting the specific type of tasks for each FSI.

In level 3 (the task selection section) paralleling and default logic has been introduced. Regardless of the answer to the first question (regarding lubrication or servicing), the next task selection question shall be asked in all cases. When following the hidden or evident safety effects paths or mission effects paths, all subsequent questions shall be asked. In the remaining nonmission economic effects paths, a "YES" answer subsequent to the first question allows exiting the logic.

A.4.4.2.2.1 Default logic. In the absence of accurate information to answer "YES" or "NO" to questions in the third level, default logic dictates a "NO" answer be given, and the

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subsequent question be asked. As "NO" answers are generated, the only choice available is the next question, which in most cases provides a more conservative, stringent, or costly route.

A.4.4.3 Procedures. Step by step instructions to guide the analyst through the RCM decision logic are provided below.

A.4.4.3.1 First level (categorization of items).

a. FSIs versus "Other Items."

Question 1: Is this item being evaluated as an FSI?

For a "YES" answer, the analyst shall proceed to question 3 (A.4.4.3.2a). For a "NO" answer, the analyst shall categorize it as "Other Item" and proceed to question 2 (A.4.4.3.1b).

b. Requirements for nonsignificant items. Question 2 shall be asked of each "Other Item".

Question 2: Is this item similar to existing items?

For a "YES" answer, task selection shall be based on past experience with like or similar equipment. For a "NO" answer, task selection shall be based on manufacturer's recommendations.

Tasks resulting from question 2 should be kept to a minimum but may include items necessary to meet minimum requirements for the comfort and morale of the operators, and maintainers. Further analysis of these items is not required.

A.4.4.3.2 Second level (consequence of failure).

a. Evident versus hidden functional failure.

Question 3: Is the occurrence of a functional failure evident to the operator during the performance of normal duties?

Question 3 shall be asked for each functional failure of the item being analyzed. The intent is to segregate the evident and the hidden functional failures.

A "YES" answer indicates the functional failure is evident; proceed to question 4 (A.4.4.3.2b). A "NO" answer indicates the functional failure is hidden; proceed to question 6 (A.4.4.3.2d).

b. Direct adverse effect on safety.

Question 4: Does the functional failure or secondary damage resulting from the functional failure have a direct adverse effect on operating safety?

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For a "YES" answer the functional failure or secondary damage resulting from functional failure has to have a direct adverse effect on operating safety.

Direct: To be direct, the functional failure has to achieve its effect by itself, not in combination with other functional failures (no redundancy exists, and it is a minimum essential equipment item).

Adverse effect on safety: This implies that the consequences are extremely serious or possibly catastrophic and might cause injury to personnel or extensive damage to equipment.

Operating: This is defined as the time interval from the moment of equipment start for the purpose of maintenance or mission to equipment shutdown.

For a "YES" answer, tasks shall be developed in accordance with A.4.4.3.3.1. For a "NO" answer, the analyst shall proceed to question 5 (A.4.4.3.2c).

c. Direct adverse effect on mission.

Question 5: Does the functional failure have a direct adverse effect on mission capability?

For a "YES" answer, task selection shall be handled in accordance with A.4.4.3.3.2. For a "NO" answer, task selection shall be handled in accordance with A.4.4.3.3.3.

d. Hidden functional failure safety effect.

Question 6: Does the combination of a hidden functional failure and one additional failure of a system or back-up function have an adverse effect on safety?

For a "YES" answer, task development shall be handled in accordance with A.4.4.3.3.4. For a "NO" answer, proceed to question 7 (A.4.4.3.2e).

e. Hidden functional failure mission effect.

Question 7: Does the combination of a hidden functional failure and one additional failure of a system or backup function have an adverse effect on mission capability?

For a "YES" answer, task selection shall be handled in accordance with A.4.4.3.3.5. For a "NO" answer, task selection shall be handled in accordance with A.4.4.3.3.6.

A.4.4.3.3 Third level (effect categories). Once the applicable second level questions are answered, the analyst is directed to one of the six effects categories:

a. Evident safety effects.

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b. Evident mission effects.

c. Evident nonmission economic effects.

d. Hidden safety effects

e. Hidden mission effects.

f. Hidden nonmission economic effects.

A.4.4.3.3.1 Evident safety effects. The evident safety effects category is to be approached with the understanding that tasks are required to assure safe operation. All questions in this category shall be asked. If no effective tasks result from this category analysis, redesign is mandatory.

A.4.4.3.3.2 Evident mission effects. The evident mission effects category is to be approached with the understanding that tasks are required to assure mission success. All questions in this category shall be asked. If' no effective tasks result from this category analysis, a redesign may be required depending on the extent of impact a failure would have on mission success (see A.4.4.2d.(l)).

A.4.4.3.3.3 Evident nonmission economic effects. The evident nonmission economic effects category indicates tasks are desirable if the cost is less than the cost of repair. Analysis of the failure modes through the logic requires the first question (lubrication or servicing) to be answered. Either a "YES" or "NO" answer to question "A" requires movement to the next level; from this point on a "YES" answer completes the analysis, and the resultant tasks satisfy the requirements. If all answers are "NO", no task has been generated. If economic penalties are severe, a redesign may be desirable (see A.4.4.2d.(2)).

A.4.4.3.3.4 Hidden function safety effects. The hidden function safety effect requires tasks to assure the availability necessary to avoid the safety effects of multiple failures. All questions shall be asked. If there are no tasks found effective, then redesign is mandatory.

A.4.4.3.3.5 Hidden function mission effects. The hidden function mission effect category requires tasks to assure the availability necessary to avoid the mission effects of multiple failures. All questions shall be asked. If no effective tasks result from this category analysis, the extent of impact a failure may have on mission success could make redesign mandatory (see A.4.4.2d.(l)).

A.4.4.3.3.6 Hidden function nonmission economic effects. The hidden function nonmission economic effect category indicates that tasks are desirable to assure the availability necessary to avoid the economic effects of multiple failures. Movement of the failure modes through the logic requires the first question (lubrication or servicing) to be answered. Either a "YES" or "NO" answer requires movement to the next level; from this point on, a "YES" answer completes the analysis, and the resultant tasks satisfy the requirements. If all answers are "NO",

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no tasks have been generated. If economic penalties are severe, a redesign may be desirable (see A.4.4.2d(2)).

A.4.4.3.3.7 Third level (task development). Task development is handled in a similar manner for each of the effect categories. For task determination, it is necessary to apply the failure modes for the functional failure to the third level of the logic diagram. There are seven possible task resultant questions in the effect categories.

a. Lubrication or servicing (all categories).

Question A: Is a lubrication or servicing task applicable and effective?

Applicability criterion: The replenishment of the consumable has to reduce the rate of functional deterioration.
Effectiveness criterion – safety: The task reduces the risk of failure.
Effectiveness criterion – mission: The task reduces the risk of failure.
Effectiveness criterion – nonmission-economic: The task is cost-effective.

b. Operator monitoring (evident functional failure categories only).

Question B: Is the ability to detect degradation of the function by normal operator monitoring applicable and effective?

Applicability criteria: Reduced resistance to failure is detectable and rate of reduction in failure resistance is predictable. Indicators that annunciate failures at the time of occurrence are not applicable.
Effectiveness criteria – safety: The monitoring is part or the normal duties of the operator and reduces the risk of failure to assure safe operation.
Effectiveness criteria – mission: The monitoring is part of the normal duties of the operator and reduces the risk of failure to assure mission success.
Effectiveness criterion – nonmission economic: The monitoring is part of the normal duties of the operator.

c. Operational check (hidden functional failure categories only).

Question B: Is a check to verify operation applicable and effective?

Applicability criterion: Verification of operation is possible.
Effectiveness criterion – safety: The task ensures availability of the hidden function to reduce the risk of multiple failures.
Effectiveness criterion – mission: The task ensures availability of the hidden function to reduce risk of multiple failures.
Effectiveness criteria – nonmission economic: The task ensures availability of the hidden function in order to avoid economic effects of multiple failures and has to be cost-effective.

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d. Inspection or functional check (all categories).

Question C: Is the ability to detect degradation of the function by on-equipment or off-equipment task(s) applicable and effective?

Applicability criteria: Reduced resistance to failure is detectable and rate of reduction in failure resistance is predictable.
Effectiveness criterion – safety: The task reduces the risk of failure to assure safe operation.
Effectiveness criterion – mission: The task reduces the risk of failure to assure mission success.
Effectiveness criterion – nonmission economic: The task is cost effective, that is, the cost of the task is less than the cost of the failure.

e. Restoration (all categories).

Question D: Is a restoration task to reduce failure rate applicable and effective?

This task includes work (on or off equipment) necessary to return the item to a specific standard.

Applicability criteria: The item shows functional degradation characteristics at an identifiable age and a large proportion of units survive to that age. It has to be possible to restore the item to a specific standard of failure resistance.
Effectiveness criterion – safety: The task reduces the risk of failure to assure safe operation.
Effectiveness criterion – mission: The task reduces the risk of failure to assure mission success.
Effectiveness criterion – nonmission economic: The task is cost effective, that is, the cost of the task is less than the cost of the failures prevented.

f. Condemn (all categories).

Question E: Is a condemn task to avoid failures or to reduce the failure rate applicable and effective?

This task includes the removal from service of an item at a specified life limit. Condemn tasks are normally applied to parts such as cartridges, canisters, engine disks, and durability structural members.

Applicability criteria: The item shows functional degradation characteristics at an identifiable age and a large proportion of units survive to that age.
Effectiveness criterion – safety: A safe-life limit reduces the risk of failure to assure safe operation.
Effectiveness criterion – mission: A safe-life limit reduces the risk of failure to assure mission success.

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Effectiveness criterion – nonmission economic: An economic-life limit is cost-effective, that is, the cost of the task is less than the cost of the failures prevented.

g. Combination (safety categories only).

Question F: Is there a task or combination of tasks which is applicable and effective?

Since this is a safety category question and a task is required, all possible avenues shall be analyzed. The most effective task or combination of tasks shall be selected.

h. Combination (mission categories only).

Question F: Is there a task or combination of tasks which is applicable and effective?

For the mission category, a review of all applicable tasks is necessary to assure mission success. From this review, the most effective task or combination of tasks shall be selected.

A.4.4.4 Setting preventive maintenance task frequencies or intervals. Determine whether real and applicable data is available which suggest an effective interval for task accomplishment. If there is no prior knowledge from similar systems or if there is insufficient similarity between the previous and current systems, the task interval or frequency can only be established initially by experienced personnel using good judgment and operating experience in concert with accurate data (reliability, redundancy, and dispatch).

A.4.5 Corrective maintenance prediction.

A.4.5.1 Corrective maintenance prediction assumptions. The following assumptions shall be used for corrective maintenance predictions:

a. Failure rates experienced are all in the same proportion to those predicted,

b. Only one failure at a time is considered,

c. Maintenance is performed in accordance with established maintenance procedures,

d. Maintenance is performed by maintainers possessing the appropriate skills and training, and

e. Only active maintenance time is addressed; administrative and logistic delays and cleanup are excluded.

A.4.5.2 Corrective maintenance elements. Corrective maintenance actions consist of the following tasks: Preparation, Fault Detection and Isolation (FD&I), and Fault Correction. Fault Correction is further broken down into Disassembly, Interchange, Reassembly, Alignment, and Checkout. The time to perform each of these tasks is an element of MTTR; the associated task

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times are MTTR elements. The definitions and abbreviations for the MTTR elements used in prediction models are shown in Table A-1.

TABLE A-I. MTTR elements.

MTTR Element (abbreviation)[1]	**Definition**
Preparation Time (PT_n)	Time associated with those tasks required to be performed before fault isolation can be executed.
Fault Isolation Time (FIT_n)	Time associated with those tasks required to isolate the fault to the level at which fault correction begins.
Disassembly Time (DT_n)	Time associated with gaining access to the replaceable item or items identified during the fault detection process.
Interchange Time (IT_n)	Time associated with the removal and replacement of a faulty replaceable item or suspected faulty item.
Reassembly Time (RT_n)	Time associated with closing up the equipment after interchange is performed.
Alignment Time (AT_n)	Time associated with aligning the system or replaceable item after a fault has been corrected.
Checkout Time (CT_n)	Time associated with the verification that a fault has been corrected and the system is operational.
Start-Up Time (ST_n)	Time associated with bringing a system up to the operation state it was in prior to failure, once a fault has been corrected and the operation status of the system verified.

The n subscript indicates that the nth replaceable item (RI) is the object of the maintenance action.

A.4.5.3 Corrective mathematical model. MTTR shall be calculated by the following equation, derived from Equation D-1 in Appendix D of MIL-HDBK-470:

[1] The abbreviations have been changed from those of MIL-HDBK-470 so as to eliminate subscripts of subscripts, which can be confusing and difficult to read, and to delete the index for multiple FD&I indications.

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$$MTTR = \frac{\sum_{n=1}^{N} \lambda_n R_n}{\sum_{n=1}^{N} \lambda_n}$$

where:

N is number of replaceable items (RI),
λ_n is the failure rate of the nth RI, and
R_n is the mean repair time of the nth RI.

(Reference MIL-HDBK-470)

This essentially says that the MTTR is the weighted average repair time for each possible repair, weighted by the failure rate corresponding to that repair.

A.4.5.4 MTTR prediction procedure. For each repairable item, estimate the failure rate and a time for each of the MTTR elements (see Table A-1), ensuring that the model and failure rate data are consistent with that of the basic reliability prediction (see A.4.2). The mean repair time for any repairable item is the sum of the times for its MTTR elements; the weighted mean repair time for the repairable item is the product of its failure rate and its mean repair time. Remember that identical repairable items used in different locations may have different mean repair times, as its accessibility may affect the disassembly, interchange, reassembly, and alignment times. Using the equation in A.4.6.3, sum the weighted repairable item mean repair times and divide by the sum of the repairable item failure rates to calculate the MTTR.

A.4.5.5 Maintenance action time standards. Task times shall be developed from the following sources in the order given:

a. Actual times experienced on the subject equipment,

b. Standard times from Tables D-IV and D-V of MIL-HDBK-470,

c. Actual times experienced on similar equipment,

d. Other recognized time sources, or

e. Engineering judgment.

A.5 TESTING PROVISIONS

A.5.1 Reliability test requirements. The reliability tests shall be conducted in accordance with the reliability test procedures which have been approved by the procuring activity. Testing shall be continued until a reject decision has been reached or the total required test time has been completed, whichever comes first.

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A.5.2 Reliability test records. Reliability test records shall be maintained as specified in the approved test procedure.

A.5.3 Performance parameter measurements. The test sample performance parameters to be measured and the frequency of measurement shall be as specified herein. When the value of any required performance parameter is not within specified limits, a failure shall be recorded. If the exact time of failure cannot be determined, the failure shall be presumed to have occurred at the time of the last recorded observation or successful measurement of that same parameter. Observations and measurements shall be made at the specified interval and recorded during the test cycle. At least one set of measurements shall be recorded when a test sample is first energized after any specified shutdown period.

A.5.4 Reliability compliance. Reliability compliance shall be reviewed by the procuring activity after each test sample failure is categorized or at any other appropriate time. Compliance shall be based on the total accumulated test time and the total number of chargeable failures at the time of the review.

A.5.5 Failure reporting, analysis, and corrective action system (FRACAS). A closed loop system shall be used to collect data, analyze, and record timely corrective action for all potential failures that occur during the preproduction and operational tests. The FRACAS shall be established and conducted in accordance with 4.5.4.

A.5.6 Classification of failures. All failures shall be classified as relevant or non-relevant. Relevant failures shall be further classified as chargeable or non-chargeable. The procuring activity will make the final determination of failure classifications.

A.5.7 Corrective maintenance demonstration.

A.5.7.1 Corrective maintenance demonstration task selection and performance. At least 30 corrective maintenance tasks shall be performed. Task selection shall be as follows.

a. Step 1. Column 1 – Identify the major units which comprise the equipment.

b. Step 2. Column 2 – Subdivide each unit to the functional level at which maintenance for the demonstration is to be performed in accordance with the approved maintenance plan. This level may be an assembly, module, printed circuit card, or piece part.

c. Step 3. Columns 3 & 4 – For each functional level of maintenance identified in Column 2, identify in Column 3 the type of maintenance task or tasks to be performed and in Column 4 the estimated mean maintenance time for the task. The maintenance tasks and estimated maintenance time would be derived from a maintenance engineering analysis, a maintainability prediction effort, or from historical data. The same maintenance task, such as "remove and replace" of a module may result from different faults within the module. Column 3 would

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identify the maintenance task and not the fault or failure which results in the occurrence of the task.

d. Step 4. Column 5 – Determine the failure rate [failures per million hours] for each module and printed circuit card, for which the maintenance task was identified in Column 3. The failure rates used should be the latest available from an associated reliability program.

e. Step 5. Column 6 – Determine the quantity of items in each major unit associated with each task in Column 3.

f. Step 6. Column 7 – Determine the duty cycle for each item associated with each task in Column 3 (such as operating time of a receiver to the operating time of the radar; engine operating hours to aircraft flight hours).

g. Step 7. Column 8 – Group together the maintenance tasks identified in Column 3 which have both:

(1) Similar maintenance actions. NOTE: A maintenance action is an element of a maintenance task. Although the estimated maintenance time for different maintenance tasks may be similar, the actions may be different, that is, one task may involve significant diagnostics, and another involve minimum diagnostics but significant access time.

(2) Similar estimated maintenance times. The maintenance times in each group should be within a range that shall not exceed the smallest value in the group by more than 50 percent.

Task grouping should be limited to within major units identified in Column 1.

h. Step 8. Column 9 – Determine the total failure rate for each task grouping identified in Column 8. The total failure rate is equal to the sum of the products of Columns 5, 6, and 7 for all tasks within the group.

i. Step 9. Column 10 – Determine the relative frequency of occurrence for each task grouping by dividing the sum of the total failure rate (sum of Column 9) into the individual total failure rate for each group.

j. Step 10. Column 11 – Fixed Sample – A sample of maintenance tasks equal to four times the sample size specified for the selected test method (see … B.4.0 of [MIL-HDBK-470]) or as specified or agreed upon with the procuring activity should be allocated among the task groups in accordance with the relative frequency of occurrence of the task group.

k. Step 11. Column 12 – The maintenance tasks to be demonstrated … are allocated among the task groups in accordance with the relative frequency of occurrence of

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maintenance for the group. The maintenance task to be demonstrated is then randomly selected from the maintenance tasks allocated to the group or modules and assemblies, within the group (Column 11). The maintenance task to be demonstrated is not returned to the sample pool and is therefore demonstrated only once.

l. Step 12. Column 1 – Identify the maintenance task of interest.

m. Step 13. Column 2 – Determine the failure modes which will result in the maintenance task of interest.

n. Step 14. Column 3 – Determine the effect of each failure mode identified in column 2.

o. Step 15. Column 4 – Determine the relative frequency of occurrence of each failure mode.

A.5.7.2 Corrective maintenance demonstration accept and reject criterion. The accept and reject criterion shall be calculated by:

$$\text{Accept if the specified maximum } MTTR \geq \overline{X_C} + \frac{\phi \hat{d}_C}{\sqrt{n_C}}$$

where:

$\overline{X_C}$ is the average corrective maintenance time for the demonstration,

φ is the value of the normal distribution function for the specified level of consumer's risk, β (see Table A-2, below), and

$\hat{d}_C$ is the standard deviation of sample of the corrective maintenance tasks.

$$\text{Reject if the specified maximum } MTTR < \overline{X_C} + \frac{\phi \hat{d}_C}{\sqrt{n_C}}.$$

TABLE A II. φ vs. **β**.

ϕ	**β (percent)**
0.84	20
1.04	15
1.28	10
1.65	5

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CONCLUDING MATERIAL

Custodians:
Army – AT
Navy – AS
Air Force – 184

Preparing activity:
Air Force – 184

Review activities:
Army – MI
DLA – GS

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